Created by: Alrayyan-A on GitHub https://github.com/Alrayyan-A

Impact of Smoking Status, Age and Region on Medical Insurance Costs

What factors have the strongest influence on medical insurance charges, and how do charges differ between smokers and non-smokers across age groups and regions?

Business Question: What factors (such as smoking status, weight, and age) have the greatest effect on medical insurance charges?
Why This Matters? Knowing this information will help reduce costs and risk, and create preventative health programs.

Data Description:

Source: “Medical Insurance Cost Dataset” - Mosap Abdel-Ghany on Kaggle
Number of records: 1338
Features (used fields):

- Age
- Smoker
- Region (in the US)
- Charges

Tools & Technologies Used:

- Microsoft Excel: Summary descriptive statistics
- MySQL: EDA, querying
- Microsoft Power BI: Searching for trends, visualizations, DAX measures

Data Cleaning & Preparation:

Raw data was checked for missing values and validated for incorrect/misspelled values. This step is important because clean and prepared data greatly affects its reliability.

Exploratory Data Analysis (EDA):

- Data is skewed to the right (mean>median>mode)
- There is a weak correlation between age and charges

Techniques used:

- Data distribution analysis. (Excel’s built-in Data Analysis → Descriptive statistics tab)
- Correlation (Excel’s built-in Data Analysis → Correlation tab)

SQL Querying:

- Aggregated ages into groups using CASE statement
- Change boolean values of Smoking to “smoker” & “non-smoker” using CASE statement
- Created a view with the modified table
- What are the average insurance charges for smokers?

  Code:
  ```
  select
  round(avg(Charges), 2) as "Average Charges"
  from medical_insurance
  where Smoker = "Yes"
  order by Charges desc
  ```
- What is the age of top 5 beneficiaries that incur the lowest charges?

  Code:
  ```
  select age, charges from medical_insurance
  order by Charges asc
  limit 5
  ```

Data Visualization and Report Creation:

- Pie chart showing the average insurance charges of smokers and non-smokers.
- Bar chart showing the average insurance charges by age groups.
- 100% stacked bar chart compared the proportion of smokers to non-smokers above 60 years old in the 4 US regions.

- DAX measurements:
- What is the average insurance charge of beneficiaries aged above 60?

Average charges 61-65 = CALCULATE(AVERAGE('Edited Insurance table ready for visualization'[Charges]), 'Edited Insurance table ready for visualization'[Ages Above or below 60] = "Above 60")

- What is the percentage difference in average charges between beneficiaries over 60 and those aged 60 or below?

% difference above 60 and below 60 = ([Average charges 61-65] - [Average charges 18-60]) / [Average charges 18-60]]

Report Goal: quick way for non-technical stakeholders to understand what factors are affecting costs the most.

Key Insights:

- Smoking status is the main driver of insurance costs.
- Insurance charges increase as age increases.
- Individuals in the northeast region incur the lowest charges.

Recommendations:

- Market to non-smokers aged 18-35 from the northeast.
- Organize preventative health programs for smokers.

Limitations:

- No information about the duration of smoking in beneficiaries
- Absence of date column to analyze the charges over time
- Lack of previous medical history

Conclusion:

A medical insurance costs data set created by Mosap Abdel-Ghany from Kaggle with 1338 records containing the following columns: age, sex, BMI, children, smoker and region was analyzed through Exploratory Data analysis on MS Excel, SQL querying on MySQL and further calculations visualizations on MS Power BI. It was found that the smoking status of the beneficiaries was the most strongly associated with high insurance charges, with old age being second. Additionally, residents of the southeast region incur the highest charges. The relevance of this analysis is determining drives of high insurance costs to reduce costs and risk and also to create helpful preventative health programs.

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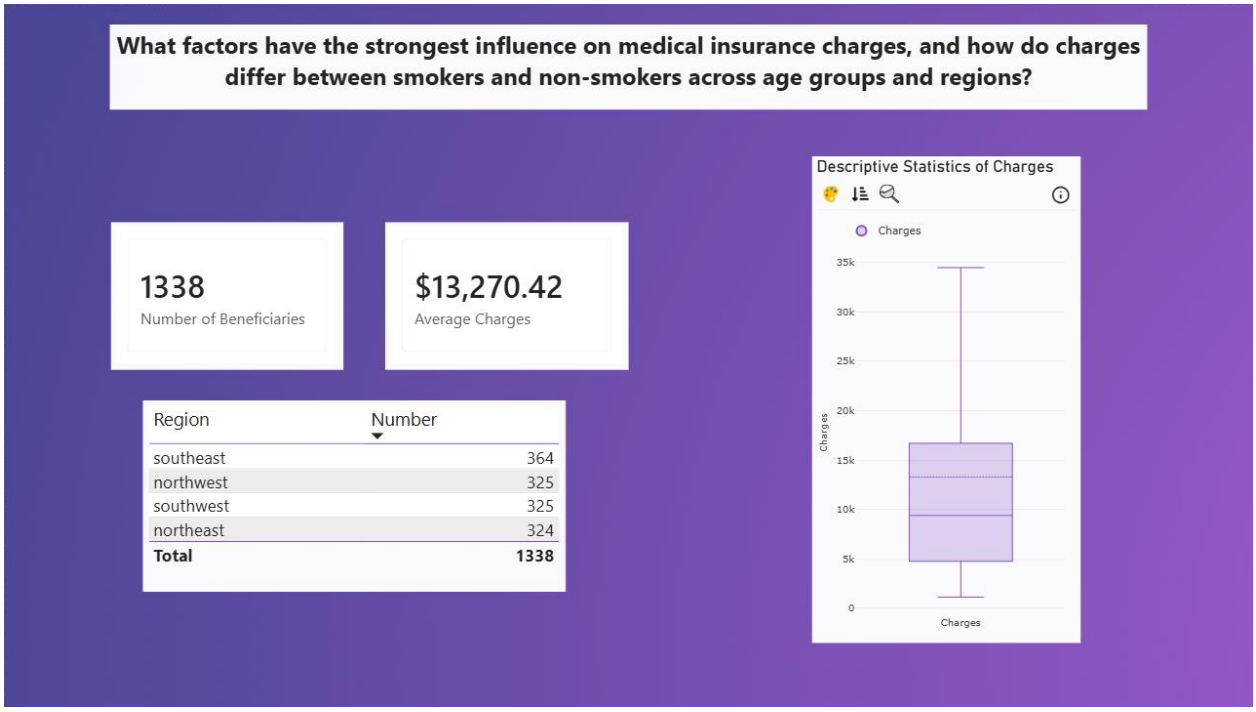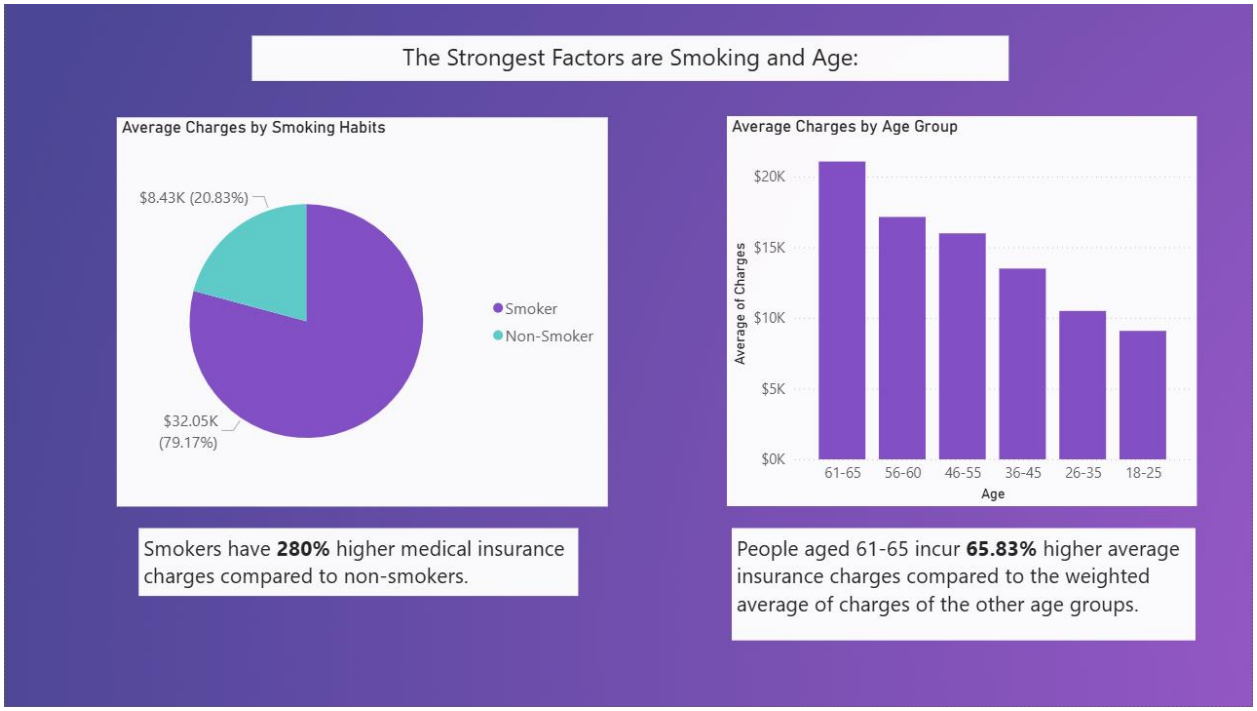
Report Screenshots: